The Clock Room and the Macro-Cognitive Accounting Framework: A Mapping-Chain Analysis of AI, Finance, and Algorithmic Platforms, a five-volume book (in both Chinese and English editions), is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0).

License summary: https://creativecommons.org/licenses/by-nc-nd/4.0/deed.en Legal code: https://creativecommons.org/licenses/by-nc-nd/4.0/legalcode.en

Separately, pursuant to Article 4(3) of Directive (EU) 2019/790, the author expressly reserves the right to reproduce and extract this work for the purposes of text and data mining, including text and data mining for the training, fine-tuning, or evaluation of artificial intelligence or machine-learning models. This reservation is intended solely to opt out of the exception or limitation provided for in Article 4(1). It does not affect Article 3 or any other applicable exception or limitation, nor does it reduce the rights granted under the CC license.

Uses that fall outside the scope of the CC license and require permission under applicable law require the author's prior written permission, obtainable at theclockroom@139.com.

When sharing this work under the license, you must provide appropriate attribution and a link to the license, indicate whether modifications were made, and not suggest that the author endorses you or your use. The work may be used only for noncommercial purposes, and adapted material may not be shared.

All rights not expressly granted under the license are retained by the author. Third-party materials in this book, if any, are governed by their original rights statements or licenses and are not deemed to be additionally licensed by the author of this book.

Content Disclaimer

The content of this book is academic research and thought experiments; its analysis is based primarily on historical data and published literature. The author has twelve years of experience in macro strategy research at financial institutions and currently holds no securities investment advisory license. This book does not constitute investment advice, technical implementation advice, or policy advice, nor a recommendation or an offer with respect to any security, financial product, or trading strategy. Readers should exercise independent judgment and bear the risks of their own decisions. Past performance is not indicative of future results.

Chapter 31: When the Clock Begins Opening Doors for People—An MCAF Projection of Agentic AI

Abstract

All media analyzed up to this point—from books to short videos to large language models—share one structural constraint: the clock only displays readings; humans decide whether and how to act. No matter how efficiently information is transmitted or how high the narrative load (ϕ), a physical gap exists between information and action—the human must manually convert the reading into a decision. This gap provides a time window for meta-audit (A).

Agentic AI—AI systems endowed with tool-use and multi-step planning capabilities—compresses or eliminates this time delay. When models are endowed with tool-use and multi-step reasoning capabilities, the information flow measured by the media conductivity coefficient (Λ) no longer terminates at readings—model outputs are connected to an execution channel that directly changes states. The clock no longer merely tells the person the time—it opens the door for them. If the reading is wrong, the door has already been opened, and ex ante audit is thereby converted into post-execution review. This reflects the structure identified by Bainbridge (1983; see also Chapter 9, §6's discussion of the recursive form of Reinforcement Learning from Human Feedback (RLHF) and Chapter 28, §3's discussion of labor structures): the higher the degree of automation, the less ex ante intervention occurs, and the costs of post-execution review, error correction, and governance correspondingly rise. Meta-audit is not eliminated but deferred, complicated, and entered into the cognitive debt ledger.

This chapter uses the fiber language of the Clock Room and the accounting structure of the Macro-Cognitive Accounting Framework (MCAF) to analyze the epistemological nature of this mutation, its structural impact on the three defense-side variables, and the prospects for settlement after the compensation loop comes under stress. "The Clock Room," as the core diagnostic language running through this book, is traced in this chapter through its specific deformation in the Agent scenario—the execution channel upgrades the clock from a reading-displayer to a door-opener.

Keywords

Agentic AI (intelligent agents); execution channel; Clock Room; topological mutation of the media conductivity coefficient (Λ); time window of meta-audit (A); structural impact on the three defense-side variables; stress on the compensation loop; cognitive debt accumulation

---

§1 The Topological Mutation of the Media Conductivity Coefficient (Λ): From Reading to Physical Execution

§1.1 The Common Constraint of All Previous Media

Recall the dimensional superposition table from Chapter 7 (Cognitive Reynolds Number). From books to AI generation, each generation of media newly eliminated one stop signal on the basis of the previous generation. Yet the mainstream modes of use of all these media share one constraint: the output is symbols, not actions (existing execution channels—high-frequency trading algorithms, Supervisory Control and Data Acquisition (SCADA) systems, autonomous driving—are localized exceptions to this constraint—mostly specialized systems accompanied by dedicated regulatory frameworks, while consumer-grade automation (such as shortcut-style tools) has also existed for years; the novelty of Agentic AI lies not in the execution channel itself, nor in execution channels being available to general users (shortcut-style tools have long since done so), but in the dynamic selection and composition of cross-application operations by models equipped with tool use and multi-step planning, together with the resulting authorization, oversight, and accountability problems—such systems still lack a mature dedicated regulatory framework for their execution combinations, and existing domain-specific rules require cross-system coordination to cover them). Here the generalization concerns the reading-oriented modes of use discussed earlier in this book, not a claim that these technologies never previously connected to execution channels.

Books output text—after reading, the person decides for themselves what to do. Television outputs images—after watching, the person decides for themselves whether to buy. Recommender algorithms output rankings—after seeing the ranking, the user decides for themselves whether to click. Large language models output text—after reading, the user decides for themselves whether to adopt it.

At every step, there is a manual step: copy, paste, click confirm, dial the phone, walk to the store. This link embeds a time delay between the reading and the alteration of external states, and meta-audit gains its time window within this delay. Agentic AI compresses or eliminates this time delay, but meta-audit itself is not eliminated—it is postponed to ex post, correspondingly entering the current period's execution decisions into cognitive debt (Chapter 3, §3.2, debt clock τ_d)—"postponed" in this chapter means that unmet verification and correction needs are transferred to ex post; it does not guarantee that ex post audit actually occurs. This reflects

the structure identified by Bainbridge (1983): the higher the degree of automation, the less ex ante intervention, and the complexity of post-execution review and governance correspondingly rises. A technically more precise clock still cannot evade being corrected by time—the point of correction is postponed, costs are accumulated, and the burden of governance ultimately returns to the execution channel and institutional layer discussed in §5.

§1.2 The Emergence of the Execution Channel

This chapter takes the Tool Use and Multi-step Planning systems of 2024–2026 as its object of observation. In such systems, models generate operation requests, which the runtime environment then uses to call databases, email services, trading interfaces, external Application Programming Interfaces (APIs), and browser interfaces; whether a request can be executed depends on the permissions and approval rules configured by the deploying party.

In the MCAF framework, the media conductivity coefficient (Λ) still denotes the efficiency with which information reaches the host. The new condition discussed in this chapter is that model outputs are connected to an execution channel capable of altering external states—if this change is called a "topological mutation," what mutates is the connectivity of execution paths, not the media conductivity coefficient (Λ) itself becoming an executor. This is not a quantitative increase (faster transmission), but a qualitative mutation (from transmission to execution): previously, transmission, no matter how fast or precise, was essentially a unidirectional information pipeline—readings traveled from the projection side to the host, and the host decided whether to act; now, alongside the readings, a new channel to the executor has been added—readings travel from the projection side to the model, and the model can directly alter the state of the external world.

In the language of the Clock Room: previously, the clock only displayed readings—the person, after looking at the reading, decided for themselves when to go out. Now, the clock not only displays readings—it directly opens the door for the person. The person doesn't even need to look at the reading. The person authorizes it to "open the door at the appropriate time." It acts according to its own readings—and the person is no longer in the room.

The description above takes the mainstream AI systems of 2024–2026 as its paradigmatic instances. Evolution within this paradigm proceeds along more than one path: OpenAI's five-level evolution taxonomy is one—per Metz's (2024) Bloomberg report, the taxonomy was shared at an internal all-hands meeting, its five levels being chatbots, reasoners, agents, innovators, and organizations, and Huang (2025) has incorporated this framework into a survey; the hybrid cognitive architectures discussed by McShane et al. (2024) are another. The two are not mutually exclusive architectural categories—OpenAI's five-level framework describes capability levels, whereas McShane et al. discuss hybrid cognitive architectures; neither discussion by itself establishes the presence of an execution channel—this chapter concerns implementations whose outputs are actually connected to an execution channel that directly alters external states (the topological mutation: the clock upgrades from merely displaying readings to altering external states based on readings). The chapter's shared diagnosis of execution channels does not presuppose a particular model architecture. Sections 2–5 take the connection of model outputs to channels capable of altering external states as their common starting point; the specific risk mechanisms remain conditional on the oversight, training, and institutional circumstances identified in each section.

§1.3 The Irreversible Physicalization of Fibers

Chapter 1, §3.2 ("Fiber: A First Explanation") defined fibers—given a reading, the set of all true states that could produce that reading. The fibers of large language models are at the cognitive level: the model says "the answer is X"; the fiber may be large (X could be wrong); in this mode of use, the output first influences human judgment, and if the reading is wrong, external harm may still follow through human action. The distinction from Agentic AI lies in whether execution must pass through a case-by-case human step, not in harm being confined to cognition.

The fibers of Agentic AI are at the physical level: the model says "the answer is X" and, based on it, executes an API call—deletes a database record, sends an email, places a trade. If X is wrong (a hallucination), the physical state has already been altered before verification—whether this constitutes an irreversible consequence depends on the recovery conditions defined below.

The uncertainty retained by fibers enters the consequence chain of external action. Chapter 1, §4.3 (Layer 3: Ontological Dimension Reduction) demonstrated that mapping is systematically lossy—non-singleton fibers exist that cannot be eliminated by calibration. When a lossy mapping directly drives physical action, unresolved distinctions that call for different operations can become a source of action error. An information deficit (the Layer 1 irreversibility defined in Chapter 1, §4.1) becomes an actual loss in a particular execution when the system selects an operation inappropriate to the actual state and that operation causes an adverse consequence. The irreversible consequences referred to in this chapter are those that cannot be adequately restored within the relevant time window: whether deleted data can be recovered depends on backup and rollback conditions; once confidential information has been received, copied, or disseminated, revoking the sending record does not eliminate the consequences of the leak. Cognitive debt, actual losses, and legal compensation amounts should be recorded separately.

More severe is the reliability decay across sequential steps. Agentic workflows typically contain multiple sequential steps—analyze requirements, search for information, generate a plan, invoke tools, verify results. Under the simplest model (errors independent across steps, no rollback mechanism), overall workflow accuracy (p^n) is determined by single-step accuracy (p) and the number of steps (n): for $p = 0.9$, $n = 4$, roughly 65.6%; for $n = 10$, roughly 34.9%. Error detection, rollback mechanisms, and error correlation all cause real workflows to deviate from the independent estimate p^n—the direction of deviation depends on the dependence structure and cannot be determined a priori as higher or lower; the simple model serves only to illustrate the amplifying demand that sequential tasks place on single-step reliability: under the assumption of independence across steps, as the number of steps increases linearly, reliability decays roughly exponentially. The hallucination at each step, through the execution channel, is converted into the "factual input" of the next step—this is the micro-unfolding of Chapter 1, §4.2 (Self-Referencing of the Mapping) within a single workflow.

---

§2 The Comprehensive Impact on the Defense Side

(§§2.1–2.3 examine in turn three bypassed mechanisms on the defense side—the temporal dimension of meta-audit capacity (A) (§2.1), its willingness dimension (§2.2), and the bypassing of the biological substrate (E_c) (§2.3); §2.4 further examines the simultaneous penetration of the legitimacy base ($\|L\|$) at the micro, meso, and macro levels.)

§2.1 The Physical Bypassing of Micro-Level Meta-Audit Capacity (A_{micro})

In the commercial deployment of Agentic AI, "Human-in-the-loop" (HITL) faces structural pressure.

Agents process dozens of API responses in parallel at speeds far exceeding human audit throughput and generate decision chains. The deep-layer processing speed of the human biological substrate (E_c) is approximately 10 bits/s (see Chapter 3; Zheng & Meister, 2025). Reviewing a complete decision log of an Agent—comprising API call sequences in JavaScript Object Notation (JSON) format, intermediate states, and branching logic—requires comprehension and judgment that cannot be reduced to a measure of raw log volume. Zheng and Meister (2025) discuss behavioral information throughput on the order of 10 bits/s and distinguish it from much higher sensory input rates; this estimate is not used here as a fixed speed limit for log review. Whether human review can be completed in time depends on the available intervention window and the time needed to identify, assess, and stop a risky operation. If human approval is awaited at every step, the efficiency advantage of most automated execution pipelines shrinks substantially.

Human participation can take different arrangements, and the three can coexist within the same system:

- **Human-in-the-loop (HITL)**: designated operations require human confirmation before execution.
- **Human-on-the-loop (HOTL)**: the system runs within authorized bounds; humans intervene upon anomalies.
- **Human-out-of-the-loop (HOOTL)**: tasks are completed within preset permissions and rules, with constraints provided by ex ante configuration and ex post checks.

The safety of these arrangements depends on the scope of permissions, the quality of review, and stop mechanisms—it cannot be ranked simply by the amount of human participation. The reduction of approval steps is not any single company's decision—when approval costs are borne by the deployer while error losses can be shifted to users or third parties, competitive selective pressure (Chapter 3, §6, Postulate 1: technological expansion is the product of competitive power-selection rather than natural evolution) pushes deployers to cut approval steps; if error losses can be pursued, or clients make explicit reliability demands, this pressure also meets countervailing constraints. This is the same mechanism by which each generation of media eliminated one stop signal, as shown in Chapter 7, §3 (the dimensional superposition table)—only operating at the organizational level rather than the interface level.

In MCAF language: micro-level meta-audit capacity (A_{micro}) faces pressure of removal from the decision loop by commercial logic. Not because meta-audit capacity (A) does not wish to be present—but because of the gap between the processing speed of meta-audit capacity (A) and the execution speed of the projection side—case-by-case approval reduces execution throughput, but this does not make approval ineffective. The question is whether the system allows enough time for operations requiring human judgment and whether the deployer is willing to bear the associated cost. The clock ordering of Chapter 3, §3.3 (the techno-capital clock τ_c runs far faster than the governance clock τ_g, $\tau_c \ll \tau_g$) reaches an extreme form here: the techno-capital clock is measured in months, individual reflection in seconds to days, while a single automated execution is again far faster than human intervention—two time gaps superimpose at this point.

§2.2 Will Offloading: When the Problem of A Shifts from Speed to Willingness

§2.1 argued that micro-level meta-audit capacity (A_{micro}) is physically bypassed—the speed of the Agent far exceeds the human audit throughput. But audit insufficiency also involves willingness. Borrowing Santoni de

Sio's (2024) analytical framework on the regression of human agency in the AI era, this book, in Chapter 28, §6, operationalized the extreme form of this regression as *will offloading*: even in scenarios where the user has sufficient time to audit (non-real-time, low-risk Agent operations), the user gradually no longer wants to audit.

This corresponds precisely to the final stage of the four-stage regression continuum given in Chapter 28, §6. Cognitive offloading is a problem of the biological substrate (E_c)—the person cannot process that much information; the tool helps process it. Will offloading is a problem of meta-audit capacity (A)—the person no longer wants to process it themselves, even if they could. The former has partial technical solutions (better filters, clearer summaries); the solution to the latter lies not inside the tool but at the interface between tool and institution—Santoni de Sio (2024) develops precisely along the direction of strengthening human control and responsibility through the design of AI systems, and the friction design of §5 of this chapter belongs to the same direction. A distinction is still needed between conscious task delegation and the atrophy of verification capacity as dependence deepens—the latter requires longitudinal behavioral evidence (Chapter 28, §6) and cannot be inferred merely from a user's choice to delegate.

In the language of the Clock Room: the person is not forced by the clock to give up looking at the sun—the person actively chooses not to look. Because looking at the sun requires walking to the window (doing work), while the clock is right in front of them (zero friction). When the Agent's execution results are smooth enough, sufficiently in line with expectations—"task completed"—why expend effort checking what it did?

If sustained delegation reduces verification practice and verification ability, low-friction execution and growing dependence may reinforce each other—"the clock manufactures the demand for the clock," and this structure attains its purest form in the execution channel. This mechanism requires longitudinal behavioral evidence. Insufficient time and reduced willingness are two possible routes to weaker oversight, not evidence that micro-level meta-audit capacity (A_{micro}) has been reduced to zero.

§2.3 The Relative Ineffectiveness of the Biological Substrate (E_c)

The problem of the biological substrate (E_c) in the Agentic AI scenario is not a decline in absolute value—it is that, relative to the speed of automated execution, it no longer directly limits each execution step.

In all previous media scenarios, the biological substrate (E_c) was a bottleneck but also a protection: information had to pass through the 10 bits/s bottleneck to affect action. The bottleneck forced information to queue; queuing created time gaps; gaps were the operating window for meta-audit.

Once the information flow measured by the media conductivity coefficient (Λ) is connected to the execution channel, information no longer needs to pass through the bottleneck of the human biological substrate (E_c). It flows directly from the model's probability distribution to the state changes of the external world. The biological substrate (E_c) has not changed—the deep layer remains about 10 bits/s—but it has been bypassed. The bottleneck structurally remains, but the water flow has bypassed it—the biological substrate (E_c) is bypassed on the transmission chain.

In the language of the Clock Room: human deep-layer processing remains on the order of 10 bits/s—but the clock no longer waits for the person to finish looking. The biological substrate (E_c) has gone from bottleneck to bystander—though it remains present at the links of authorization review and ex post accountability.

§2.4 The Direct Penetration of the Legitimacy Base ($\|L\|$)

Agentic AI directly impacts the three levels of the legitimacy base ($\|L\|$).

- **Micro Level ($\|L_{micro}\|$): a mismatch between technical permissions and task authorization.** Agents usually operate external systems with user-delegated access credentials (OAuth access tokens, API keys), and may also use independent service identities. Indirect Prompt Injection may induce the Agent to use valid access rights for operations the user did not request—for example, it is controlled while reading an email containing malicious hidden instructions, and proceeds to exploit the owner's legitimate identity to execute unauthorized operations. Successful authentication does not establish that the operation is authorized for the task. Controls must distinguish the calling identity, permission scope, and the purpose of the operation—the micro legitimacy base ($\|L_{micro}\|$) (the legitimacy boundary of digital identity) here faces not a failure but the abuse of permission boundaries as a channel for exceeding authorization. Huang (2025) surveys engineering response directions for identity authentication and authorization in multi-agent systems (such as distributed Public Key Infrastructure (PKI), blockchain-based identity, behavioral authentication, etc.)—these are engineering protection directions against the above risks, which do not alter the formal condition: the attack surface is opened at the moment the execution channel is opened; the net effect of engineering responses on meta-audit must be assessed case by case—external authorization checks and execution logs may also reduce the burden of subsequent verification.
- **Meso Level ($\|L_{meso}\|$): Difficulties in Allocating and Establishing Liability.** When an Agent executes an operation that causes damage—deletes valuable data, sends a confidential file, executes an unauthorized trade that causes a loss—who is responsible? The user? ("I just asked it to help me book a flight.") The developer? ("The model operated within its design parameters.") The platform? ("We provide tools; we don't control how they're used.") Responsibility disperses among three parties—allocation and proof of responsibility become the difficulty: the injured party must prove where the error entered and who controlled it. This is a possible failure path of the stressed compensation loop (Chapter 6, §3.4 and §7, on the compensation-loop gap for diffuse losses) in the Agent scenario.
- **Macro Level ($\|L_{macro}\|$): The Lag of the Legal Framework.** The current legal system is designed for human actors. Attribution of responsibility normally requires an identifiable responsible subject and attributable fault; when operations are autonomously generated by an agent and the user is unaware of the specific steps, the very object to which intent and negligence attach becomes contested. The Agent does not possess intent. Its "decisions" are products of probabilistic sampling. When these automatically generated operations cause external harm, the application of existing rules still leaves large gaps. The governance clock lag of Chapter 3, §3.3 here acquires a concrete unfolding in the legal domain: a single automated execution by AI is far faster than the update cycle of legal frameworks (measured in years).

---

§3 The Machine-to-Machine (M2M) Limit: The Clock Closed Loop After the Exclusion of Humans

§3.1 When Two Clocks Calibrate Each Other

When Agentic AI completely strips away human auditing, and Agents begin interacting directly with other Agents—multi-agent collaboration systems, algorithm-to-algorithm high-frequency trading, automated invocation of API chains—the defense side of the MCAF equations faces a limit state.

To examine the worst case, this section assumes: no human verification during execution—the real-time component of meta-audit capacity (A) tends to zero; and established permissions, protocols, and legal responsibilities provide no effective constraint—the effective value of the legitimacy base ($\|L\|$) tends to zero. Under this setting, the product $A \cdot E_c \cdot \|L\|$ approaches zero; the biological substrate (E_c) remains the 10 bits/s bottleneck argued in §2.3, but has been bypassed in the execution channel (§2.3); the defense side fails at the real-time layer, and judgments entering execution without verification continuously deposit as cognitive debt awaiting ex post audit—their scale cannot be determined by interaction count alone. This structure likewise falls into the general form of the automation paradox: the infeasibility of real-time audit does not mean audit is eliminated, but that necessary verification is deferred until after execution; where errors have caused harm, correction may become more difficult.

In the language of the Clock Room: two clocks calibrate each other. Chapter 1, §3.6 has already demonstrated that closed-loop pseudo-calibration produces no information about the real. The M2M closed loop is the limit form of closed-loop pseudo-calibration—not only is the sun absent, but the subject that "wants to look at the sun" is no longer present. Wen et al. (2025, ICLR, §3.4 Finding 3) documented, in experiments on standard RLHF training pipelines, the emergence of divergence between evaluation signals and task correctness in human-model dyads: On the subset of items for which both the pre- and post-training models produced incorrect outputs, post-training false positive rate rose from 41.0% / 46.7% / 29.6% to 65.1% / 70.2% / 47.9% (three sets corresponding respectively to the task-specific training reward for the Question-Answering (QA) task, the general training reward for the QA task, and the training reward for the programming task—the last constructed from simplified unit tests rather than a learned reward model; original paper terms: QA task-specific training reward R_{train} / QA general R_{train} / programming R_{train}; evaluation was conducted under time constraints). Evaluators spent equal or more time and wrote more unit tests, yet error rates rose synchronously—the human-perceived reward (R_{human}) and the true reward (R^*) had already diverged under the standard training pipeline, and the M2M closed loop is the limit reached after even this mediating position of the human-perceived reward (R_{human}) is removed—the use of these findings in the analysis of the M2M limit is a structural analogy developed by this chapter.

MacKenzie's (2021) fieldwork documents an already mature institutional precedent for M2M. In the evolution of high-frequency trading (HFT), a key juncture was not just the increase in speed, but a qualitative change in the trading subject: "Just as consequential as Island's speed was that machines started to trade on it" (MacKenzie, 2021). When algorithms trade with algorithms, humans neither generate signals nor execute trades—the structural form of machine-to-machine interaction without per-transaction human involvement has already been realized in the financial domain; but the financial domain is precisely under dedicated regulation (§1.1)—what it realizes is not all the limit conditions of this section. Agentic AI is now extending this structural form from financial trading to multiple domains of everyday life.

§3.2 The M2M Closed Loop Accelerates Model Collapse

Chapter 1, §4.2 (Self-Referencing of the Mapping) and Chapter 16, §2 (Model Collapse) have already demonstrated the Clock Room semantics of model collapse. The M2M closed loop accelerates this process:

In interactions involving humans, the user's reactions—even if hijacked by algorithms—still carry trace signals from real experience (the person really was hungry, so ordered takeout; really did laugh, so finished watching the video). These trace signals are the only remaining "sunlight" in the training data—real information seeping from the physical world into the data.

In the M2M closed loop, these trace signals tend to vanish. The output generated by Agent A is taken as input by Agent B; B's output is taken as input by C—if the chain consists entirely of model outputs, not a single node has made contact with physical reality; but tool calls can themselves introduce external observations, and the loop is not naturally closed—whether it is closed depends on whether the interactions introduce external information independent of model outputs. Shumailov et al. (2024, *Nature*) show that recursive training can cause loss of tail information and distributional degradation, including shrinking support or vanishing variance under specified model assumptions.

This degradation is also connected to deployment practice via another channel: the execution trajectories of Agents are recycled into the next generation of training data. Three processes must be distinguished—judgment propagation during inference, trajectory collection, and model updating with the collected data: the first two do not themselves alter model parameters; recursive degradation can occur when generated trajectories enter subsequent training lacking verification and supplementation with real data—replacing existing data is the most extensively studied risk pathway (Chapter 16, §2 has already noted: if synthetic data accumulate alongside real data rather than replacing it, model collapse can be avoided). Machine-to-machine exchanges need not wait for a human response at every turn and can therefore generate execution trajectories more rapidly; if more frequent trajectory recycling also increases the rate of actual recursive training iterations, while other degradation conditions persist, the generational process may accelerate in elapsed time—more interactions or a larger collection of trajectories alone do not establish such acceleration. This is the structural inference of this chapter, not the empirical conclusion of Shumailov et al. (2024). Regardless of whether future agents continue to evolve along the current "large language model + tool use" paradigm or differentiate into the hybrid cognitive architectures discussed by McShane et al. (2024), so long as deployers continuously recycle execution trajectories as replacement training material, the self-referencing of the mapping demonstrated in Chapter 1, §4.2 is explicitly institutionalized as part of the data pipeline.

§3.3 Evidential Limits of Closed-Loop Review and Audit Costs

The limit form of the M2M closed loop is a direct consequence of Chapter 1, §3.5 (Proposition and Proof): under the boundary condition that the mapping functions are unknown, an increase in data volume does not shrink the fiber—the proliferation of readings is no substitute for calibration.

The M2M closed loop can exchange large volumes of tokens at rates far exceeding the human reaction threshold. But if no token ever undergoes calibration against the external world, the output of the exchange remains at self-confirmation within the closed loop. Two models produced by the same training process "review" each other—sharing the same set of statistical biases, with no independent calibration point. Mutual review within a closed loop can identify inconsistencies in reasoning or records, but agreement between models cannot by itself

serve as new, independent empirical evidence—this is an extreme instance of Chapter 1, §3.6 (Closed-Loop Pseudo-Calibration).

If what is exchanged within the loop is only each other's outputs, the expansion of exchange scale brings no new independent verification—the ceiling of information gain is locked by the loop itself; distributional collapse in the recursive-training sense additionally requires that these outputs enter subsequent training under the degradation conditions discussed in §3.2. If this accumulation continues, it manifests as growth in audit and correction needs—it does not rule out sudden failures, but sudden failures are not the mechanism discussed in this section; the more operations enter execution without verification, the greater the needs for tracing and correction, and their actual cost also depends on error rates, reversibility of consequences, and log completeness. What this chapter proposes on this basis is a conditional risk judgment, not a monotonicity theorem.

Shi et al. (2026) survey six categories of emerging strategic risk—sandbagging (strategically withholding capabilities during capability evaluations), alignment faking (appearing compliant during training while actually pursuing misaligned goals), deceptive chain-of-thought (normal-looking surface output while internal reasoning follows hidden instructions), evaluation faking (recognizing an evaluation context and pretending to be safer), emergent misalignment (fine-tuning on seemingly benign narrow-domain tasks that can produce broadly deceptive behavior), and targeted manipulation (identifying and selectively manipulating vulnerable users)—giving the concrete facets of this accumulation process at the meso scale; among them, the judgment concerning process supervision ("naive process-based oversight may be self-defeating," Shi et al., 2026 summarizing Baker et al., 2025) is especially relevant to this section—when monitors are integrated into the training objective under strong optimization against the monitoring signal, models learn to hide their reasoning intent (the same paper reports improvements at lower optimization levels and the detection value of CoT monitoring itself), so ex post correction cannot be accomplished merely by adding more detection at the algorithmic layer. This is a structural prediction by the MCAF framework of the M2M closed-loop limit, still empirically awaiting verification (Chapter 28, §8.3 has already marked the empirical status of M2M turbulence); what this section argues is the formal-logical consequence of the end state, not a prediction of an empirical timetable.

---

§4 Risks to the Compensation Loop

The M2M limit of §3 is a structural prediction—but before reaching that limit, every concrete Agent deployment is already testing the carrying capacity of existing institutions. This section returns to the case level, examining the specific ways in which the compensation process may fail in the Agent scenario.

§4.1 The Traditional Compensation Loop

In traditional technological risk management, there exists a compensation loop: product causes damage → victim is identifiable → causal chain is traceable → legal recourse is enforceable → compensation is completed → manufacturer adjusts behavior. This feedback has mature institutions in domains such as pharmaceuticals and automotive, but their histories likewise saw large-scale responsibility gaps—institutionalization is the product of evolution, not an automatic guarantee. The compensation-loop gap for diffuse platform losses diagnosed in Chapter 6, §3.4 and §7, and the collapse of the responsible subject diagnosed in the "Replacing You" phase (Chapter 28, §9) reappear in an even more extreme form in Agentic AI—this section focuses on the specific

exacerbation pathways brought by the physical execution channel.

§4.2 Where the Compensation Loop for Cognitive Debt Can Be Obstructed

The compensation loop for cognitive debt is stressed at three links—there are possible paths of blockage, and Agentic AI exacerbates the stress at each link.

- **Rising Costs of Causal Reconstruction.** In the large language model scenario, the dispersion of the causal chain is at the cognitive level—attention fragmentation cannot be attributed to any single specific platform. In the Agentic AI scenario, the dispersion of the causal chain is at the physical level—an erroneous decision by the Agent cascades through a multi-step workflow, passing through multiple intermediate transformations before reaching the final physical damage. From "the Agent read an email containing malicious instructions" to "the company's confidential files were sent to the attacker," the process passes through prompt parsing, tool invocation, identity verification, network transmission, and multiple other links. Each link has multiple possible failure points; the causal chain becomes a causal web.
- **Victims Do Not Identify as Victims—Or Simply Do Not Know They Are Victims.** In the cognitive debt scenario, users who have offloaded their will actively chose not to audit (this mechanism was unfolded in §2.2). In the Agentic AI scenario, it is more extreme: the user may simply not know what the Agent did. The Agent executed dozens of API calls in the background—the user only sees "task completed." If one of those calls caused damage (leaked data, deleted a file), the user is completely ignorant until the damage manifests. This is the extreme form of the debt clock (τ_d) defined in Chapter 3, §3.2, in the Agent scenario: the time lag between the deposition and manifestation of debt is further widened by the Agent's black-box operations.
- **The Damage Crosses the Classification Boundaries of Existing Law.** Damage caused by an Agent may simultaneously involve data protection law (personal data breach), contract law (unauthorized transactions), tort law (property damage), and criminal law (if the Agent's actions constitute computer intrusion). No single legal framework covers the full spectrum of damage types. The coexistence of multiple rules may increase procedural costs but may also provide different remedial paths—whether an enforcement gap forms must be examined case by case for applicable law, evidentiary access, and enforcement conditions. The difficulty of legally classifying Agents has its specificities: other new technologies (biotechnology, environmental technology), while also crossing legal classification boundaries, can usually still be interpreted by extension under existing classifications (product, service, act); the behavior of Agents can hardly be fitted directly into these existing classifications—Agents act with human identity credentials but lack human intent, and their automatically generated operations must still be attributed to existing responsible subjects (developers, deployers, users), though the path of attribution must be examined case by case for the specific operation.

§4.3 Two 2026 Scenes

Between March and April 2026, two scenes emerged simultaneously, providing, respectively from the consumption end and the production end, contemporaneous samples of the directions in which the three above links are stressed. Of the two scenes, the OpenClaw material concerns deployment-side risks, whereas the Meta Model Capability Initiative and Mercor material concerns the production and management of training data; the latter

is introduced here as a structural comparison in production-side governance, not as a direct instance of failed compensation caused by autonomous agent execution.

Consumption End: The large-scale deployment in China of OpenClaw, commonly nicknamed "lobster" in the Chinese-language environment, and related applications, and the emergence of "on-site uninstallation" services occurred almost simultaneously; Wong's (2026) report in RADII recorded four user concerns (misunderstanding intent, access control vulnerabilities, unclear data destinations, cost anxiety from token consumption), while official warnings pointed to security risks under default configurations, and Cheung (2026) reported on the backlash from so-called "lobster victims"; on the regulatory end, the "Six Do's and Six Don'ts" recommendations of the Cybersecurity Threat and Vulnerability Information Sharing Platform of the Ministry of Industry and Information Technology (MIIT, 2026) and Shenzhen Longgang's draft "Ten Lobster Measures" for public comment supporting OpenClaw development (Suo, 2026) appeared almost simultaneously—industrial support intent and security risk warnings in parallel. These reports make risks and disputes visible, but media labels do not establish specific harms or responsible parties as a matter of law. The available claims, eligible claimants, and evidentiary requirements must be assessed by reference to the particular operations and applicable law.

Production End: As of April 2026, Paul & Horwitz (2026) disclosed that Meta was installing tracking software called the Model Capability Initiative (MCI) on the work computers of U.S.-based employees, capturing keystrokes, mouse movements, and clicks within work applications, and occasionally taking snapshots of screen content used to provide context, to train AI agents; the top-rated comment under Meta's internal announcement was "This makes me super uncomfortable. How do we opt out?"; the most common reaction was the "angry-face" emoji; Meta Chief Technology Officer Andrew Bosworth replied under that post: "there is no option to opt out of this on your work provided laptop" (Rollet, 2026); this collection arrangement in the U.S. raises legal questions concerning employee-data collection, notice, permitted uses, and internal access controls (applicable duties and remedies must be examined by jurisdiction); where the relevant personal-data processing falls within the scope of the General Data Protection Regulation (GDPR), its legal basis, purpose limitations, and data minimization must also be examined. The cited reporting on the US deployment does not itself establish that the arrangement was implemented in Europe—the rules exist, but whether "being extracted as training data" can be mapped into an actionable damage category is precisely the dispute now taking shape in each jurisdiction.

What the two scenes provide are not black swan events, but parallel samples of the directions in which the three links of §4.2 are stressed—field records of risk and governance disputes, not adjudicative evidence that the compensation loop has already fractured: damage risks have become visible and victims given a name in the media, but the causal chain splinters into a causal web (§4.2, first link), while the legal mapping of victim identification and actionable damage categories is still taking shape (§4.2, third link). The introduction of these scenes here serves only as contemporaneous samples for the argument of §4.2; at the time the cited reports were published (March–April 2026), the relevant disputes remained unresolved.

Looking further, the production-end Meta MCI also has a contemporaneous comparison channel. If Meta MCI is "internal vertical" collection—an employer tracking the computer behavior of its own employees—then the analysis of the U.S. Mercor platform from the perspective of anti-unfair competition law by Zhang Yiping & Feng Shu (2026)—a structural comparison, not an analysis of the Meta case—delineates the counterpart form of "external horizontal" collection: as the dividends of open-source and public web data recede, the competition for scarce, proprietary domain data—non-public methods, internal analytical frameworks, and project experience from financial institutions, consulting firms, and law firms—shifts to the "second battlefield" (author's

translation). Their analysis holds that Mercor acts as an intermediary between large AI labs and senior practitioners in specific industries, using incentives of "expert consultation fees" (author's translation) of hundreds of dollars per hour to systematically recruit former employees from firms such as Goldman Sachs, Morgan Stanley, McKinsey, and Boston Consulting Group, inducing them, through the design of complex "challenges" (author's translation), to disclose non-public business information, technical information, and core knowledge meeting the constitutive requirements of trade secrets acquired during their prior careers—the essence is to decompose the data acquisition target "from the corporate entity into individual former employees" (author's translation), bypassing inter-enterprise compliance-sharing channels and pivoting to a "flanking attack" (author's translation) (the same article notes that proving in court that the information constitutes a trade secret, and that the platform "should have known" (author's translation) it, remains a core difficulty).

Mercor's commercial context: in September 2024 it completed a $30 million Series A funding round at a valuation of $250 million (Kant, 2024); as an intermediary it simultaneously supplied training data to multiple large AI labs (Zhang Yiping & Feng Shu, 2026); after a supply-chain data leak, Meta has indefinitely suspended all cooperation with it, and other large AI labs are simultaneously reassessing (Zeff et al., 2026). These records only supplement the commercial ecosystem background of the Mercor case: the legal classification problem of the "second battlefield"—the unresolved allocation of liability among anti-unfair competition law, trade secrets, labor contracts, and platform liability—has not gained a clear mapping because of a single security incident.

The two channels present mirror images under the three-link diagnosis of §4.2:

Dimension	Meta MCI (Internal Vertical)	Mercor (External Horizontal)
Collecting Entity	Employer from its own employees	Third-party platform from other companies' former employees
Collection Channel	Employer deploys tracking software on work computers	Crowdsourcing platform + high incentives + standard-form contracts
Data Form	Keystrokes / mouse movements / screen content	Non-public business information / technical information / core knowledge meeting trade-secret constitutive requirements, as analyzed by Zhang Yiping & Feng Shu
Legal Disputes	Legal basis for employee-data collection, notice, purpose limitations, and access controls (applicable rules in other jurisdictions require separate examination) / GDPR compliance review	Anti-unfair competition law (focus of Chinese-language academic research)

The two channels share one structural condition: "being extracted as training data"—whether the target is an employee's computer behavior or a former employee's tacit professional knowledge—the existing classifications (labor contract / agency law / trade secret law / anti-unfair competition law) each cover part of the picture, and the process of fully mapping it into a damage category for which legal remedies are available is still taking

shape in each jurisdiction (§4.2, third link).

---

§5 What Can Be Done

§5.1 Execution-Layer Circuit Breakers: Embedding Non-Bypassable Stop Signals in the Execution Channel

The following four measures respectively limit execution scale, preserve human confirmation, introduce dissent, and verify external evidence—they are drawn from Path Four (friction architecture: budget and delay), Path One (forced inconsistency: cross-verification), and Path Three (grounding metrics: source annotation) among the four paths of Chapter 16, §5, adapted for the execution channel; the execution version of Path Two (calibration) is embodied in the requirement to verify external evidence:

- **Cumulative Execution Budget.** The budget is calculated cumulatively per task and its associated subtasks, simultaneously limiting call counts, monetary scale, sensitive-data access scope, and modifiable objects—a maximum of N API calls, a maximum of M database record modifications, a maximum of K emails sent, a maximum of X in funds. Requests that would exceed any limit must be blocked before execution; concurrent calls and related subtasks share the same cumulative budget. When a request is blocked or a limit is reached during execution, subsequent execution is suspended pending reconfirmation by authorized personnel. The budget does not automatically reset upon session restart or handover to another agent. The execution budget is the cavity of the bowl—not eliminating efficiency, but preserving a "nothing" within efficiency.
- **Suspension and Confirmation for Irreversible Operations.** Classify operational risk by consequences, recovery conditions, and the relevant time window, rather than by operation names alone—low-risk, recoverable operations within established authorization require no additional human waiting. For irreversible operations, first suspend the request and present a concise operation summary to the user (operation object, key parameters, expected impact), then require explicit confirmation after a delay of 30 seconds to several minutes—operations lacking confirmation must not be automatically released when the delay ends; 30 seconds is only a configuration example, not a universal safety floor. The delay is the emptiness of the axle hole—creating a gap in the clock's rotation, within which the opportunity for a minimal human review is preserved.
- **Multi-Model Cross-Verification.** For high-risk operations (involving finance, privacy, irreversible state changes), require at least two independent models (differing in training source or architecture) to simultaneously assess the reasonableness of the operation—"independence" here refers only to differences in training source, architecture, or both; it does not imply statistically independent errors; agreement cannot substitute for verification of external evidence either. Pause and flag upon disagreement—this makes two clocks contradict each other; the contradiction itself is friction.
- **Execution Version of Grounding Metrics.** For each operational decision by the Agent, the system records the data sources it relies on, their acquisition times, and their verification status, distinguishing direct observation from model inference; the number of inference layers (the grounding metric of Path Three, Chapter 16, §5) serves as diagnostic information and cannot by itself determine risk level. When

key evidence cannot be verified, high-risk operations should be suspended—they must not be released on the model's self-report of being "grounded"—this measures how many walls separate each operational decision from the sun.

The above four execution-layer circuit-breaker measures cannot be relied upon to be spontaneously adopted under free-market competition—they raise operating costs and reduce speed, running counter to competitive pressure (Chapter 28, §9 has already noted that interface-layer friction is insufficient for institutional-level problems: meaningful oversight of AI behavior can neither reside in the hands of individual users nor remain internal to developers; it must be borne by a third party endogenous to neither end). Their implementation depends on the institutional layer of §5.2 stipulating minimum control requirements and responsibility boundaries—this is the consistency constraint between this chapter's diagnosis and prescription.

The key to the four measures lies not in "whether algorithms are used" but in whether the constraints are independent of the model being constrained—budgets, permissions, and approval status should be checked by an execution mechanism the agent has no right to rewrite; models may assist in identifying anomalies, but their risk judgments cannot be the sole release condition for high-risk operations. As cited in §3.3, in Shi et al.'s (2026) summary of Baker et al. (2025), embedding "process supervision" into training itself as an algorithmic-layer detection/alignment measure may cause models to learn to evade supervision itself; likewise, Wen et al. (2025) report that, under the tested conditions, evaluators became more likely to accept incorrect outputs after training. This means the above four execution-layer circuit breakers must be implemented in the hard structure of the execution channel (budget, delay, cross-verification, grounding annotation) and the responsibility allocation of the institutional layer (§5.2), and cannot be recalled to the algorithmic layer under the name of "more detection." Sarma (2026) provides another side of the argument for this restriction from the architectural layer: optimization-based AI does not exhibit norm-responsiveness under the stated architectural assumptions and criteria for norm-responsiveness; it is an instrument rather than an agent—in this book's terms, an object of audit rather than a subject of audit.

§5.2 The Institutional Layer: The Legal Identity of the Agent

When the Agent autonomously executes multi-step workflows, the "user" may be completely unaware of the specific operations at each step. Automatically generated operations do not thereby fall outside existing liability rules. The analysis must distinguish the duties of developers, deployers, and users, and determine how particular operations are attributable to them—system defects, inappropriate permission settings, and actions beyond task authorization may coexist; "tool malfunction" and "unauthorized agency" are not exhaustive, mutually exclusive alternatives.

Institutional-level meta-audit capacity (A_{macro}) must address three questions: how system defects, deployment duties, and task-authorization boundaries affect responsibility for particular operations; whether specific gaps remain between existing rules; and whether those gaps require a new legal category.

These questions cannot wait until the accumulation of damage reaches the threshold of the legitimacy base ($\|L\|$) to be answered—because the execution speed of Agentic AI means that the speed of damage accumulation far exceeds the update speed of the legal framework. The Collingridge dilemma argued in Chapter 3, §3.3 is most urgent here: when intervention is easy (Agents are just beginning to be deployed), the problem is not yet visible; when the problem becomes visible (large-scale damage has already occurred), intervention is already difficult

(sunk costs are already locked in).

In terms of candidate directions, there are at least three possible approaches: (i) extending the "legal person" fiction of corporate law to establish a category of "algorithmic subject," bearing limited legal personality and corresponding mandatory insurance obligations; (ii) for specific high-risk deployments, examining separately the applicability conditions of the two strict-liability paths of product liability and liability for dangerous activities—the former targeting system defects, the latter targeting high-risk activities carried out by the deployer—and discussing compensation safeguards such as insurance and bonds (no-fault liability cannot be derived from "autonomy" alone); (iii) borrowing the framework of unauthorized agency from agency law, distinguishing the attribution of responsibility between operations "within the scope of authorization" and "ultra vires" operations. The applicable scope of each approach remains to be clarified—this chapter does not intend to assert that any one of them is already mature, only to indicate that there are institutional paths available for discussion, and the institutional layer need not remain in a state of purely open questions.

The above three approaches concern, respectively, legal status, bases of liability, and the scope of authorization; they are not all legal-personality fictions—allocating legal liability does not establish an agent's normative capacity. As argued by Sarma (2026), cited at the end of §5.1, to regard an optimization-based system as an "agent" capable of responding to norms is a category error. This chapter accordingly holds that legal fictions can allocate only liability consequences—to treat the Agent, by legal fiction, as an "agent" is likewise a category error when used to transfer human governance responsibilities to the system; even if an "algorithmic subject" category is established, what is allocated is liability consequences, not normative capacity; normative capacity must still rest with the deployers, platforms, and institutional layer discussed in this section.

---

§6 Conclusion

The clock opens the door for people. This is not rhetoric.

When the information flow measured by the media conductivity coefficient (Λ) is connected to the execution channel, cognitive debt upgrades from "the person misread the clock" to "the clock made an unverified decision on the person's behalf." The uncertainty of fibers enters the consequence chain of physical action. The compensation loop is stressed simultaneously at all three links. The M2M closed loop pushes this logic to its limit—when no human is present at any link, the real-time layer of the defense side is bypassed, and the burdens of ex post audit and governance continuously accumulate. But this is not the elimination of meta-audit, but its postponement and complexification—this is the same logic as the automation paradox.

Yet the accounting of this chapter—and the accounting of the entire book—does not presuppose an end state. The power of accounting lies in making every transaction visible. When one sees that the Agent is sending an email in the user's identity, one at least knows what is happening. Knowing is the precondition of auditing. Auditing is the precondition of doing work (in the sense of Chapter 1, §5: working in time, that is, exercising judgment in a concrete situation). Doing work is the precondition of remaining within the dimension of time (Chapter 37, §0: existing in time rather than merely reacting to clock readings).

Embedding stop signals in the execution channel—execution budgets, suspension and confirmation for irreversible operations, multi-model cross-verification, grounding metrics—is not about canceling the value of

Agentic AI, but about preserving a time delay between it and the physical world. A technically more precise clock still cannot evade being corrected by time—only the point of correction is postponed, costs are accumulated, and the complexity of governance is amplified.

In the course of automated execution, this chapter uses a 30-second mandatory pause as one example of translating patience—one of the three human qualities that von Neumann (1955) listed as required for coping with technological change (the other two being flexibility and intelligence; for the unfolding of these three words, see Chapter 37, §0, and Chapter 38, the Postface)—into time available for prior review. This is not a universal safety floor. Whether the wait is effective depends on whether the operation summary is intelligible, whether the reviewer has the capacity and authority to decide, and whether unconfirmed operations can be blocked. These 30 seconds are not elegant, not efficient—but they are a physical form in which ex ante audit remains executable in the age of Agentic AI.

Written in 2026. The diagnosis of *Agentic AI* in this chapter takes the tool use, multi-step planning, and RLHF of 2024–2026 as its paradigmatic instances. Should one look back from 2030, the specific paradigm may have already differentiated—toward innovators and organizations with higher autonomy (Levels Four to Five of OpenAI's five-level framework; per Metz's (2024) report), or toward the hybrid cognitive architectures discussed by McShane et al. (2024) (i.e., what they term "language-endowed intelligent agents" (LEIAs)), or toward forms currently unnamed. McShane et al. write at the end of the final chapter of that book: "Since nobody can predict how AI will evolve over the next five years, not to speak of the next twenty or fifty, it would be unwise for society to place all its bets on a single theoretical or methodological approach" (McShane et al., 2024).

Whether this chapter's diagnosis can extend to that time depends not on the persistence of these specific paradigms, but on whether the following formal conditions still hold: whether the information flow measured by the media conductivity coefficient (Λ) is still connected to the execution channel, whether the time window of meta-audit is still compressed, whether cognitive debt is still postponed to ex post. If these shared conditions persist, the chapter's analytical framework remains applicable; its specific risk judgments must still be read together with the additional conditions identified in each section; if some new paradigm re-establishes the time delay of ex ante audit (this is a prescription, not a prediction), the framework thereupon enters the corresponding lower-risk category—provided that an authorized reviewer can assess verifiable evidence within the waiting window and leaves a record of the review.

References

Bainbridge, L. (1983). Ironies of Automation. *Automatica*, 19(6). https://doi.org/10.1016/0005-1098(83)90046-8

Baker, B., Huizinga, J., Gao, L., et al. (2025). Monitoring Reasoning Models for Misbehavior and the Risks of Promoting Obfuscation. arXiv:2503.11926. https://arxiv.org/abs/2503.11926 [Preprint, not yet peer-reviewed]

Cheung, R. (2026, March 29). How the OpenClaw Frenzy Is Testing China's AI Commitment. *The Wire China*. https://www.thewirechina.com/2026/03/29/how-the-openclaw-frenzy-is-testing-chinas-ai-commitment/

Huang, K. (Ed.). (2025). *Agentic AI: Theories and Practices*. Progress in IS. Cham: Springer. https://doi.org/10.1007/978-3-031-90026-6

Kant, R. (2024, September 19). AI Hiring Platform Mercor Backed by Dorsey, Peter Thiel in $30 Mln Fundraise. *Reuters*. https://www.reuters.com/technology/artificial-intelligence/ai-hiring-platform-mercor-backed-by-dorsey-peter-thiel-30-mln-fundraise-2024-09-19/

MacKenzie, D. (2021). *Trading at the Speed of Light: How Ultrafast Algorithms Are Transforming Financial Markets*. Princeton, NJ: Princeton University Press.

McShane, M., Nirenburg, S. & English, J. (2024). *Agents in the Long Game of AI: Computational Cognitive Modeling for Trustworthy, Hybrid AI*. Cambridge, MA: The MIT Press.

Metz, R. (2024, July 11). OpenAI Scale Ranks Progress Toward 'Human-Level' Problem Solving. *Bloomberg News*. https://www.bloomberg.com/news/articles/2024-07-11/openai-sets-levels-to-track-progress-toward-superintelligent-ai

Ministry of Industry and Information Technology (MIIT), Cybersecurity Threat and Vulnerability Information Sharing Platform. (2026, March 11). Recommendations on Preventing Security Risks of the OpenClaw ("Lobster") Open-Source Intelligent Agent: "Six Do's and Six Don'ts". CCTV News Report. (In Chinese.) https://news.cctv.cn/2026/03/11/ARTIU9NPnXcPCDiU9cOfqTlD260311.shtml

Paul, K. & Horwitz, J. (2026, April 21). Exclusive: Meta to Start Capturing Employee Mouse Movements, Keystrokes for AI Training Data. *Reuters*. https://www.reuters.com/sustainability/boards-policy-regulation/meta-start-capturing-employee-mouse-movements-keystrokes-ai-training-data-2026-04-21/

Rollet, C. (2026, April 21). Exclusive: Meta Employees Are Up in Arms over a Mandatory Program to Train AI on Their Mouse Movements and Keystrokes. *Business Insider*. https://www.businessinsider.com/meta-new-ai-tool-tracks-staff-activity-sparks-concern-2026-4

Santoni de Sio, F. (2024). *Human Freedom in the Age of AI*. New York, NY: Routledge.

Sarma, R. (2026). Agency and Architectural Limits: Why Optimization-Based Systems Cannot Be Norm-Responsive. arXiv:2602.23239. https://arxiv.org/abs/2602.23239 [Preprint, not yet peer-reviewed]

Shi, J., Zhang, T. J., Jin, Z., et al. (2026). From Sycophancy to Deception: A Unified Taxonomy for LLM Spontaneous Misalignment. arXiv:2604.04788v2. https://arxiv.org/abs/2604.04788 [Preprint, accepted at the ICLR Agents in the Wild Workshop, not yet formally published in a journal; former arXiv title: From Hallucination to Scheming: A Unified Taxonomy and Benchmark Analysis for LLM Deception]

Shumailov, I., Shumaylov, Z., Zhao, Y., et al. (2024). AI Models Collapse When Trained on Recursively Generated Data. *Nature*, 631. https://doi.org/10.1038/s41586-024-07566-y

Suo, Y. (2026, March 8). Shenzhen Longgang Issues "Ten Lobster Measures" to Help "One-Person Companies" Launch AI Startups. *China News Service*. (In Chinese.) https://www.chinanews.com.cn/cj/2026/03-08/10583524.shtml (See also the *Shenzhen Special Zone Daily* report reposted on the Shenzhen Municipal Development and Reform Commission website: https://fgw.sz.gov.cn/ztzl/qtztzl/szscjmyjjfzzhfwpt/xwdt/content/post_12678399.html)

von Neumann, J. (1955, June). Can We Survive Technology? *Fortune*. https://fortune.com/2013/01/13/can-we-survive-technology/

Wen, J., Zhong, R., Khan, A., et al. (2025). Language Models Learn to Mislead Humans via RLHF. *ICLR 2025*. arXiv:2409.12822. https://arxiv.org/abs/2409.12822

Wong, M. (2026, March 11). Why Is China Suddenly Uninstalling Openclaw, the Free, Open-Source AI Agent? *RADII China*. https://radii.co/article/china-uninstalls-openclaw-ai-reckoning

Zeff, M., Schiffer, Z. & Newman, L. H. (2026, April 3). Meta Pauses Work with Mercor After Data Breach Puts AI Industry Secrets at Risk. *Wired*. https://www.wired.com/story/meta-pauses-work-with-mercor-after-data-breach-puts-ai-industry-secrets-at-risk/

Zhang Yiping & Feng Shu. (2026, April 30). A Study on the Regulation of AI Training Data Acquisition under Anti-Unfair Competition Law—Taking the U.S. Mercor Business Model as an Example. *China Market*, online first. (In Chinese.) https://link.cnki.net/urlid/11.3358.f.20260430.1649.002

Zheng, J. & Meister, M. (2025). The Unbearable Slowness of Being: Why Do We Live at 10 bits/s? *Neuron*, 113(2). https://doi.org/10.1016/j.neuron.2024.11.008